Diffusion models

The forward and backward process, training, and conditioning

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Which are real? Which are fake?



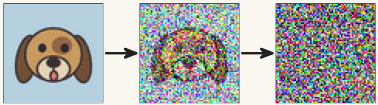
Diffusion models in brief

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Training

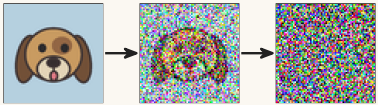


Take real images, add noise in many small steps, and learn to predict the noise that was added.

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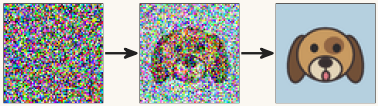
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Generation

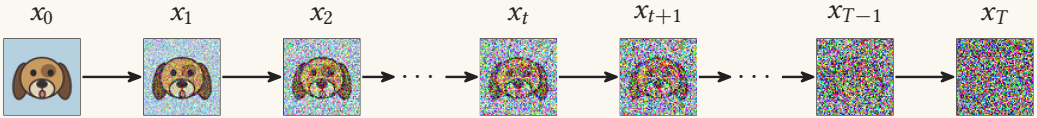


Start from pure noise and remove the predicted noise, step by step, until a clean image appears.

Forward & backward process

Each forward step fades the image and adds noise; generation runs the chain backwards.

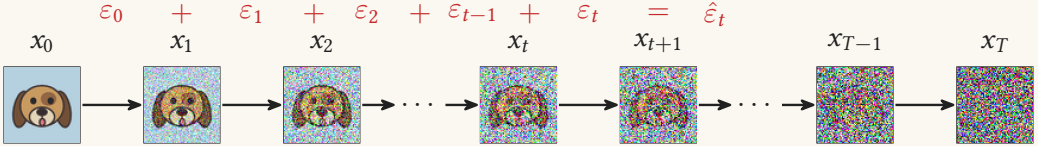
fade & noise



Forward & backward process

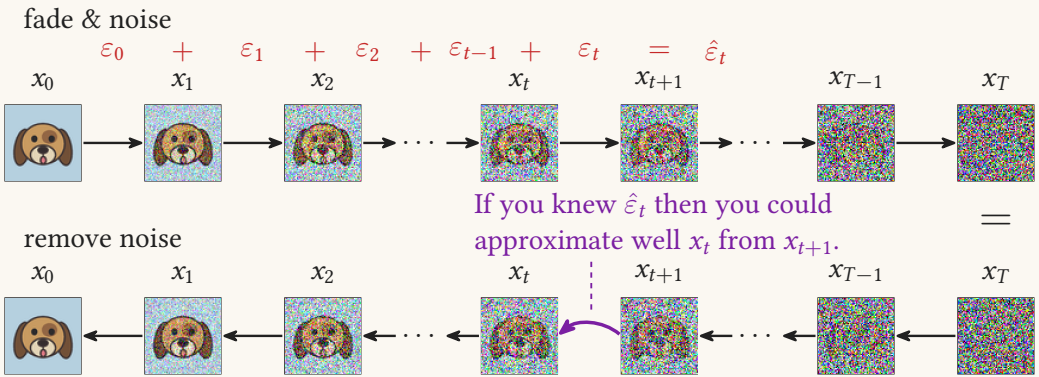
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Training of a diffusion model

The model is trained to predict the noise $\hat{\varepsilon}_t$ from the noisy image x_t.

1. Take an image of the dataset.

x_0



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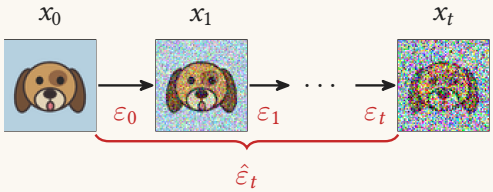
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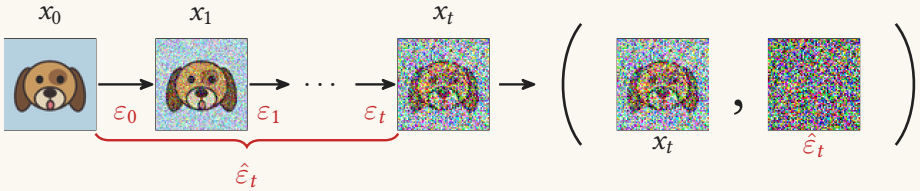
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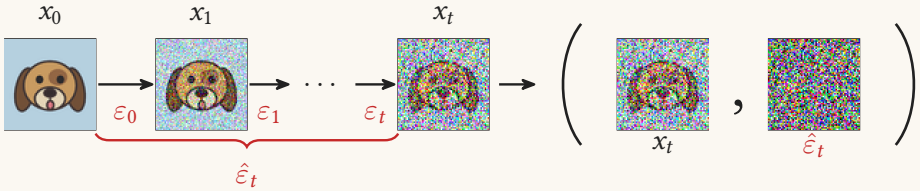
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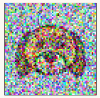


Repeat until you collect $\{(x_t^{(1)}, \hat{\varepsilon}_t^{(1)}), \ldots, (x_t^{(n)}, \hat{\varepsilon}_t^{(n)})\}$

What the model learns

One neural network, two uses: trained to predict the noise, then used to remove it.

Training



noisy image x_t

model
neural network

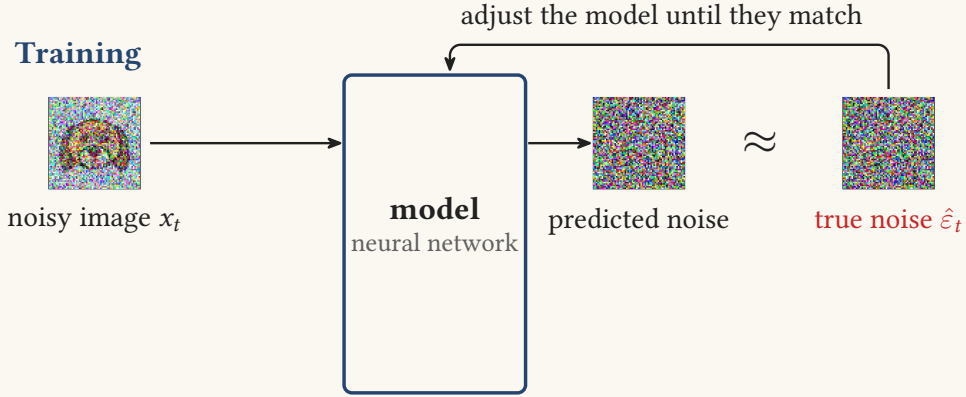
predicted noise

$\approx$

true noise $\hat{\varepsilon}_t$

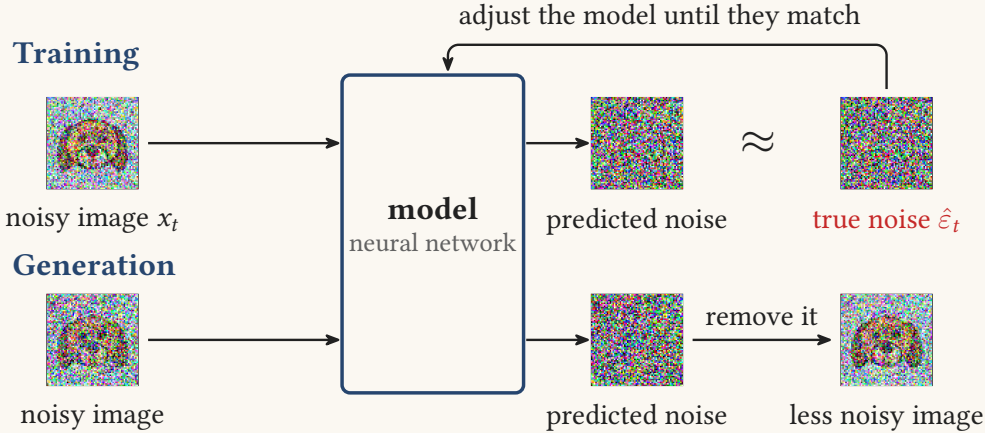
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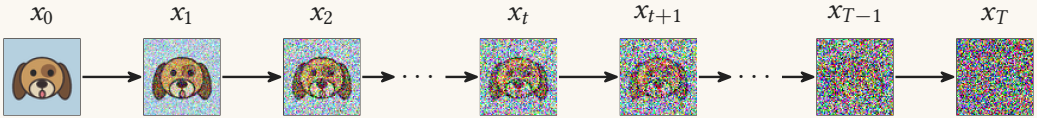


Conditional diffusion models

The prompt "cat" is given to every backward step, so the image it produces is a cat.

fade & noise

$\varepsilon_0 \quad + \quad \varepsilon_1 \quad + \quad \varepsilon_2 \quad + \quad \varepsilon_{t-1} \quad + \quad \varepsilon_t \quad = \quad \hat{\varepsilon}_t$

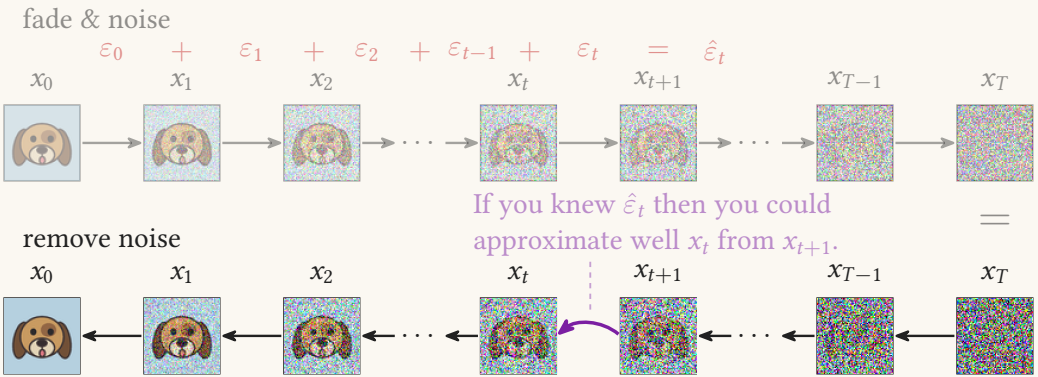


If you knew $\hat{\varepsilon}_t$ then you could approximate well x_t from x_{t+1}.

=

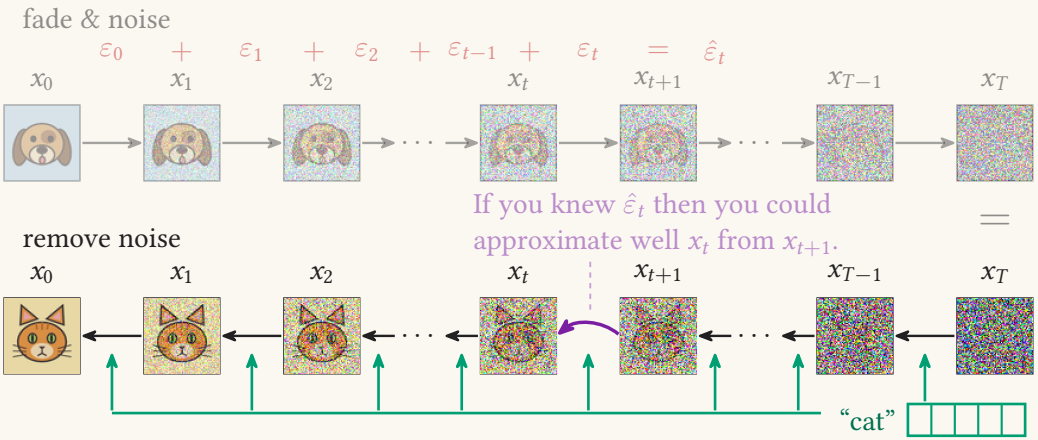
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Each training image comes with its caption, which is fed to the model as well.

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x_0



"dog"

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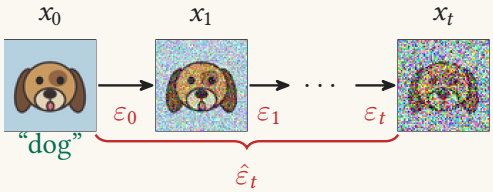


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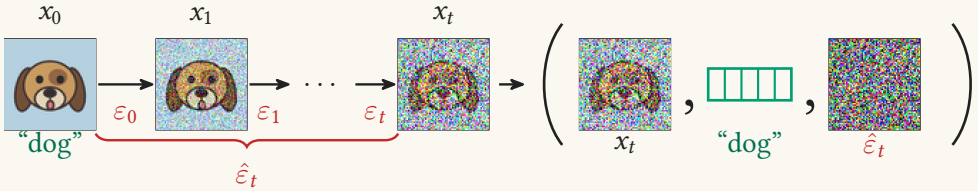
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Repeat for many captioned images.